NAME

mbpy_grd3dplot − Create an executable shellscript which will generate a GMT modern-mode 3D perspective plot of gridded data.

VERSION

Version 5.0

SYNOPSIS

mbpy_grd3dplot **−I***grid* [**−O***root*] [**−−format***fmt*] [**−−title***text*] [**−−colorbar−label***text*] [**−−dpi***dpi*] [**−X**] [**−V**] [**−R***xmin/xmax/ymin/ymax*] [**−J***spec*] [**−P***pagesize*] [**−−portrait** | **−−landscape**] [**−−scale−location***{l,r,t,b}*] [**−−tick−info***spec*] [**−−length−scale***spec*] [**−E***azimuth/elevation*] [**−F***exaggeration*] [**−−zscale***val*] [**−−color−mode***{1-7}*] [**−W***spec*] [**−D**] [**−Y**] [**−−stretch−color**] [**−Z***zmin/zmax*] [**−−data−scale***factor*] [**−−magnitude***val*] [**−−azimuth***deg*] [**−−elevation***deg*] [**−−shade−flip**] [**−−intensity−file***grid*] [**−−stretch−shade**] [**−−mesh−contour−pen***pen*] [**−−drape−file***grid*] [**−−mesh−null**[*level/r/g/b*]] [**−−mesh−zlevel***val*] [**−−coast**] [**−−coast−resolution***{f,h,i,l,c}*] [**−−coast−dry−fill***color*] [**−−coast−wet−fill***color*] [**−−coast−lake−fill***color*] [**−−coast−river***color*] [**−−coast−boundary***spec*] [**−−coast−pen***pen*] [**−−text***lon/lat/size/angle/font/justify/text*] [**−−gmt−default***param/value*]

DESCRIPTION

mbpy_grd3dplot is a Python/PyGMT-era replacement for the Perl macro **mbm_grd3dplot**. It reads a grid and generates an executable shellscript of **GMT modern mode** commands, built around **gmt grdview**, which produces a 3D perspective view of that grid when run. Several styles of view can be generated: color fill, color fill shaded by synthetic illumination, color fill shaded by a supplied intensity grid, color fill of slope magnitude, color fill shaded by slope magnitude, a plain wireframe mesh, and a wireframe mesh with contour lines.

mbpy_grd3dplot shares its page-layout, projection-fitting, and color-palette conventions with **mbpy_grdplot** (and, in fact, its underlying Python modules **mbpy_layout.py**/**mbpy_color.py**/ **mbpy_misc.py**), so a plot's color scheme and overall sizing behave the same way across both tools. What is specific to a 3D view is the viewpoint (**−−view**, azimuth/elevation of the observer) and the vertical exaggeration/z-scale (**−−exaggeration** or **−−zscale**): the effective footprint of the plot on the page is corrected for the chosen viewing azimuth, and only two of the four possible vertical frame walls (whichever face the viewer) are annotated.

Unlike **mbm_grd3dplot**'s single-letter **−M***sub***:***sub***:**... grammar, **mbpy_grd3dplot** exposes each concept as its own descriptive long option (see OPTIONS below).

mbpy_grd3dplot does not expose a **−−contour** option: **mbm_grd3dplot** itself defines a **−C***contour_control* option, but that value is never actually passed to any GMT plotting command in the original macro -- it is a no-op. The one real contour capability **mbm_grd3dplot** has is drawing contour lines directly on a mesh plot via **−−color−mode** 7's **−−mesh−contour−pen**.

Note: on at least GMT 6.7.0, **−−color−mode** 7 (mesh with contour lines) can trigger a segmentation fault inside **gmt grdview** itself when a 3D perspective and **−Jz** vertical scaling are both active together with a contour pen (**−Qm −W***pen*) -- this reproduces regardless of the pen syntax used, and appears to be an upstream GMT bug rather than anything **mbpy_grd3dplot** can control or work around. Plain mesh plots (**−−color−mode** 6, no contour pen) are unaffected.

mbpy_grd3dplot requires **GMT** 6 or later (modern mode) and a Python 3 interpreter. It depends on the same sibling support modules as **mbpy_grdplot** (**mbpy_layout.py**, **mbpy_color.py**, **mbpy_misc.py**) plus its own **mbpy_grd3dplot_script.py**, installed alongside it.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−I, **−−input** *grid*
Input grid file. Required.

−O, **−−output** *root*
Output root name. Default: the input grid's basename with its extension removed.

−−format *fmt*
Output format passed to **gmt figure**. Default **pdf**.

−−title *text*
Plot title.

−−colorbar−label *text*
Label for the color scale bar. Default **z**.

−−dpi *dpi*
Rendering resolution: both **grdview**'s own **−Qi***dpi* (**−−color−mode** 1-5 only) and the final raster's resolution.

−X, **−−execute**
Run the generated script immediately after writing it.

−V, **−−verbose**
Print the generated script's path and a one-line summary.

−R, **−−region** *xmin/xmax/ymin/ymax*
Plot region. Default: the grid's own bounds.

−J, **−−projection** *spec*
GMT map projection/scale spec. Default: an auto-fit Mercator (geographic grids) or linear (all others) projection chosen to fill the page, adjusted for the chosen **−−view** azimuth's apparent footprint.

−P, **−−pagesize** *size*
Page size code. Default **a**.

−−portrait, **−−landscape**
Force the page orientation instead of choosing automatically.

−−scale−location *{l,r,t,b}*
Color scale bar placement. Default **b**.

−−tick−info *spec*
Raw **−B** basemap tick-interval spec (covering x, y, and z axes together), overriding the automatically computed one.

−−length−scale *spec*
A GMT distance (map) scale bar spec, passed as **−L***spec* on **gmt basemap**.

−E, **−−view** *azimuth/elevation*
Viewpoint azimuth/elevation of the 3D perspective, e.g. **150/15**. Default **240/30**. Also determines which two of the four vertical frame walls are drawn/annotated.

−F, **−−exaggeration** *factor*
Vertical exaggeration. If neither this nor **−−zscale** is given, the map z-scale defaults to **2.0/(zmax−zmin)**.

−−zscale *val*
Explicit map z-scale (**−Jz***val*), overriding **−−exaggeration**.

−−color−mode *{1-7}*
1 plain color fill; 2 color shaded relief; 3 color shaded by a supplied intensity grid; 4 color fill of slope magnitude; 5 color shaded by slope magnitude; 6 wireframe mesh plot (no color fill); 7 wireframe mesh plot with contour lines (see the GMT-bug note above). Default 1.

−W, **−−color−control** *spec*
Either *style*[*/palette*[*/ncolors*]], or the path to an existing **.cpt** file to use as-is.

−D, **−−color−flip**
Reverse the color palette.

−Y, **−−no−nice−interval**
Use a direct linear color stretch instead of a rounded interval.

−−stretch−color
Histogram-equalize the color stretch (**−−color−mode** 1 only).

−Z, **−−zbounds** *zmin/zmax*
Override the grid's own detected z-range, affecting the color stretch, the vertical axis, and the z-scale/exaggeration calculation.

−−data−scale *factor*
Rescale grid values by *factor* before colorizing/shading.

−−magnitude *val***, −−azimuth** *deg***, −−elevation** *deg*
Illumination/slope-shading parameters for **−−color−mode** 2-5 (the sun's azimuth/elevation for synthetic illumination, distinct from **−−view**'s viewpoint azimuth/elevation).

−−shade−flip
Reverse the shading direction (negates the effective magnitude).

−−intensity−file *grid*
Intensity grid for **−−color−mode** 3.

−−stretch−shade
Histogram-equalize the intensity file first (**−−color−mode** 3 only).

−−mesh−contour−pen *pen*
Draw contour lines directly on the mesh (**−−color−mode** 7 only). See the GMT-bug note above.

−−drape−file *grid*
A grid to drape on the relief (**grdview**'s **−G**) instead of coloring the surface directly from the CPT.

−−mesh−null *[level/r/g/b]*
Null-plane level/color for undefined (NaN) nodes (**grdview**'s **−N**). If given with no value, defaults to <*zmin*>**/200/200/200**.

−−mesh−zlevel *val*
Z level for the base plane and frame (**grdview**'s and **gmt basemap**'s **−Z**).

−−coast
Enable a coastline overlay (**gmt coast**), projected into the same 3D perspective.

−−coast−resolution *{f,h,i,l,c}*
Coastline database resolution: full, high, intermediate, low, crude.

−−coast−dry−fill, **−−coast−wet−fill**, **−−coast−lake−fill** *color*
Fill colors for land, water, and lakes.

−−coast−river *color*
Draw rivers in the given color.

−−coast−boundary *spec*
Draw a national boundary; repeatable.

−−coast−pen *pen*
Pen for the coastline itself.

−−text *lon/lat/size/angle/font/justify/text*
Place a text label (**gmt text**), projected into the 3D view. Repeatable.

−−gmt−default *param/value*
Override a GMT default parameter for this plot; repeatable.

EXAMPLES

A shaded-relief 3D perspective view, viewed from azimuth 150, elevation 15 degrees, rendered directly to a 300 dpi PNG:

```
mbpy_grd3dplot −I topo.grd −O topo3d \
  −−color−mode 2 −−magnitude 0.2 −−azimuth 90 −−elevation 5 \
  −−view 150/15 \
  −−title "Survey Area" −−colorbar−label "Topography (meters)" \
  −−length−scale "fx10/2/32.70/5.0+lkm" \
  −−pagesize c −−dpi 300 −−format png −−execute −V
```

A plain wireframe mesh view:

```
mbpy_grd3dplot −I topo.grd −O topomesh \
  −−color−mode 6 −−view 210/25 −−title "Survey Area" \
  −−format png −−execute −V
```

SEE ALSO

mbm_grd3dplot(1), **mbpy_grdplot**(1), **mbpy_histplot**(1), **mbpy_xyplot**(1), **grdview**(1), **mbdefaults**(1), **mbgrid**(1), **mbsystem**(1)

BUGS

Please report any bugs to the MB-System project on GitHub (https://github.com/dwcaress/MB-System).